

Lecture 2: Structured Data

CS1090A — Introduction to Data Science

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Lecture outline

1. What data are, and where they come from
2. Collecting data responsibly
3. Types of variables, and how to summarize them
4. Tables: messy and tidy
5. Live: student survey exploration (notebook)



The five steps from Lecture 1. **Today is steps 2 and 3.**

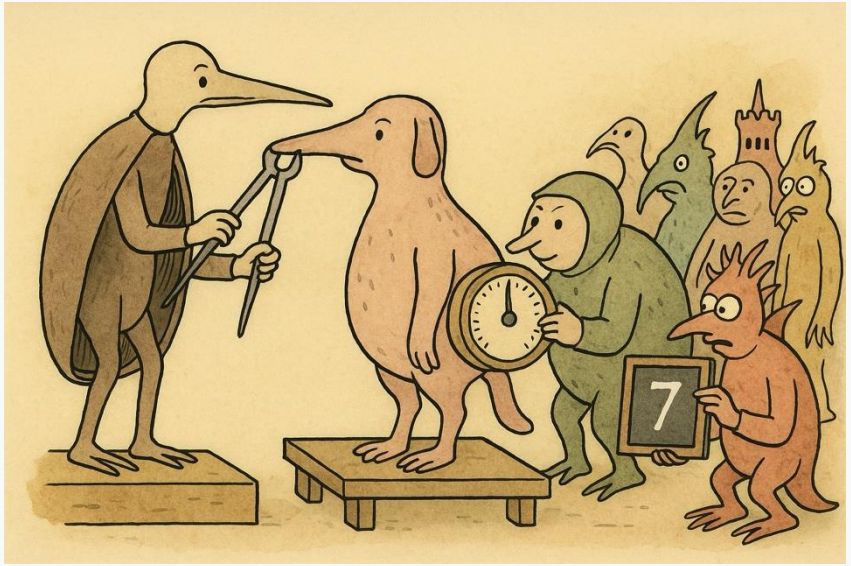
SECTION 1

What data are, and where they come from

1. **What data are, and where they come from**
2. Collecting data responsibly
3. Types of variables, and how to summarize them
4. Tables: messy and tidy
5. Live: student survey exploration (notebook)

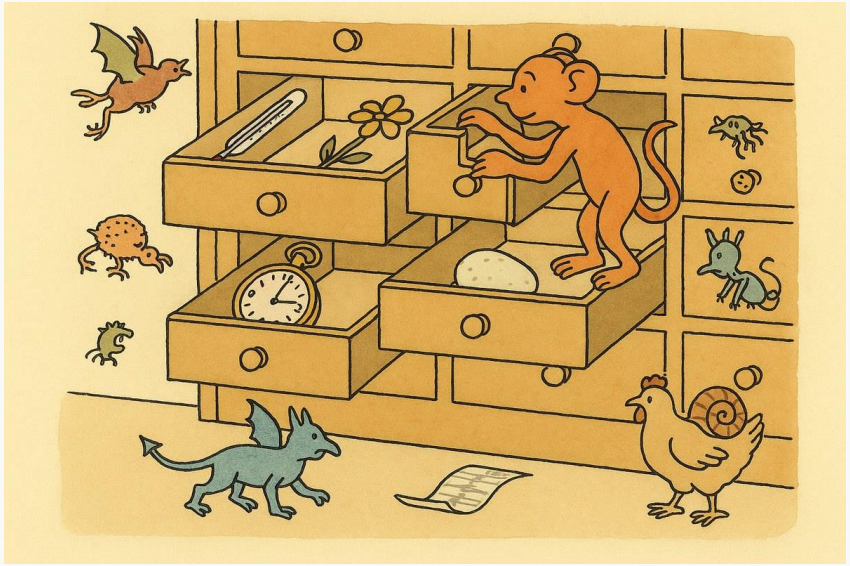
What are data?

- Possible 3-part definition:
  - an observed **entity**
  - a **variable** recorded about it
  - and that variable's **value**
- It is data only when a reader can tell **what was recorded, and on what scale**
- e.g., *This student* · *their birth month* · **2004-11**
- **Datum** is singular, **data** are plural



Where data come from

- **Secondary** — records that already exist, held by you or published by someone else
- **Primary** — *you* record them yourself: a survey, an experiment, a sensor
- HW1 collects from pages that already exist. The survey today is primary.



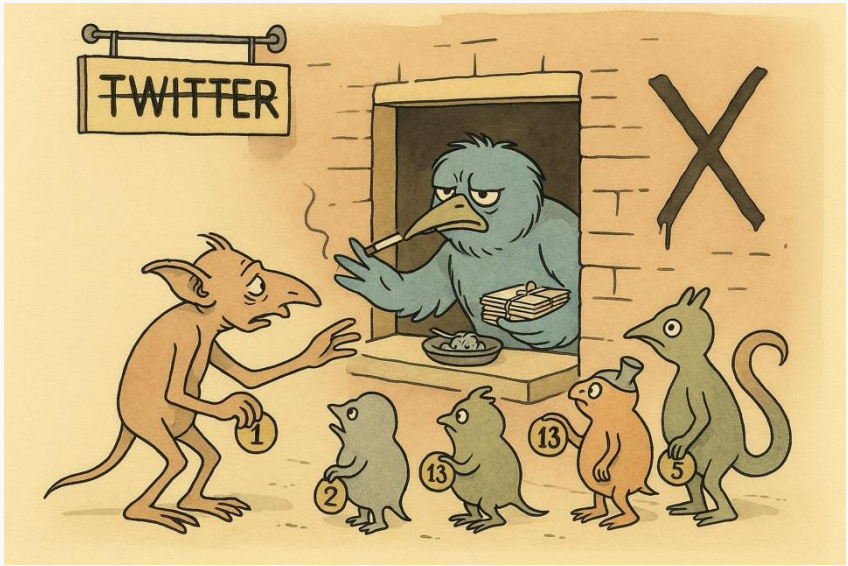
Ways to gather data that are online

- **Scraping** — take the page a browser would show, pull the data out of its HTML
- You did this in **Section 1**, and you are doing it in **HW1**
- **An API** — the site publishes a way for programs to ask directly (you also do this in **HW1**)



What using an API is actually like

- You ask a **URL** for data and get back **JSON**, not a web page
- **An API key** should be private! Never paste one into a notebook you share.
- **A rate limit** caps requests per minute. Over it, you are refused, not queued.
- **A library** usually exists — either official or a ‘wrapper' someone wrote
- Some are free, some bill per request. **Find out before your first loop, not after! $$$**



SECTION 2

Collecting data responsibly

1. What data are, and where they come from
2. **Collecting data responsibly**
3. Types of variables, and how to summarize them
4. Tables: messy and tidy
5. Live: student survey exploration (notebook)

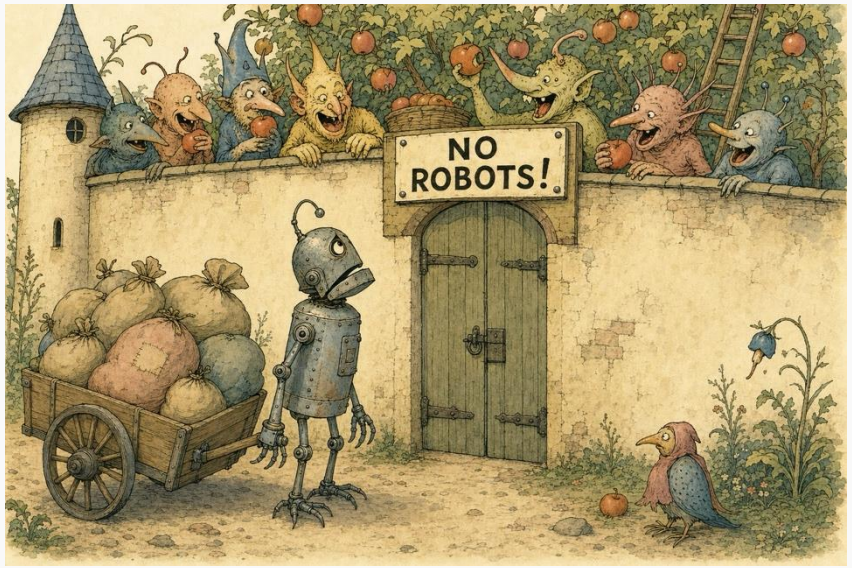
Rules for reusing publicly accessible data

- Readable in a browser ≠ **licensed for reuse**
- Example: A news photo is visible to everyone and still owned by somebody
- **A license** says what you *may* do: copy it, change it, pass it on, publish it, etc.
- Looking at a page grants none of those. Check the license before you republish.



Three questions before you collect

- **Are you allowed to?** — check the site's terms of service and/or license
- **Are you bypassing something?** — a paywall, a login, a rate limit
- **Are you causing harm?** — consider server load and the privacy of people in the data!
- `robots.txt` asks crawlers which paths to leave alone. It is a request, *not* a license!



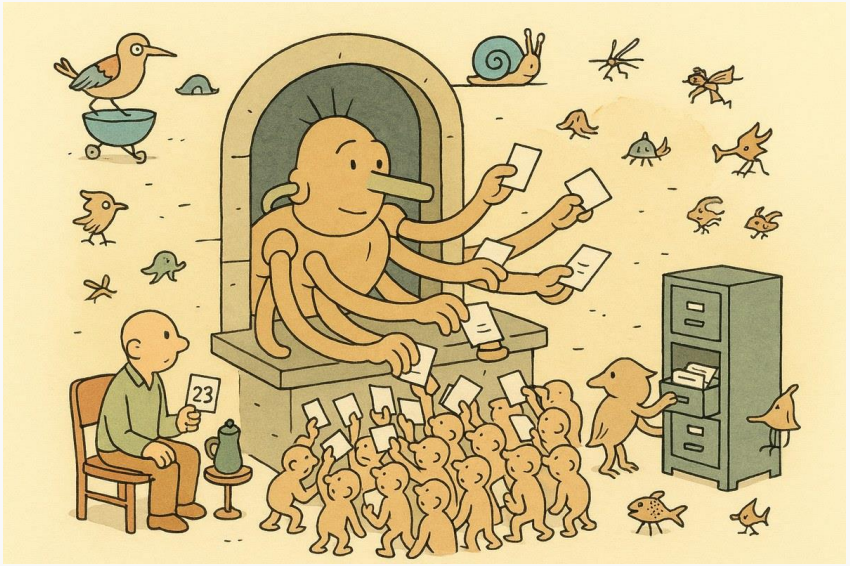
Personal data, even when they are public

- Data about **people** carry obligations that data about weather do not
- **Direct identifiers** — a name, an email, a photograph
- **Re-identification** — two datasets that name nobody on their own can name people once joined on what they share
- Being published **does not establish consent** to your analysis



Reducing the load your scraper puts on a server

- You are using **someone else's computer**
- Slow your request rate. **A 429 response (or outright blocking) means stop**
- **Cache** what you fetch — ask *once*, not on every rerun
- Identify your scraper honestly



Who is accountable for what gets collected?

- Code that runs is **not evidence that you were *allowed* to collect the data**
- Your AI agent is not (yet) a member of the moral community, nor can it bear legal responsibility.
- ***You* are accountable for the code you run and the data it collects.**



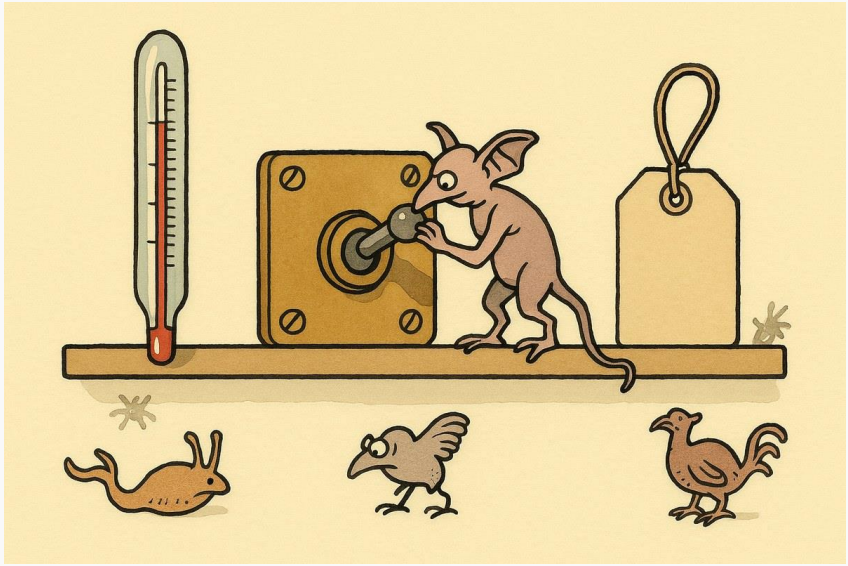
SECTION 3

Types of variables, and how to summarize them

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2. Collecting data responsibly
3. **Types of variables, and how to summarize them**
4. Tables: messy and tidy
5. Live: student survey exploration (notebook)

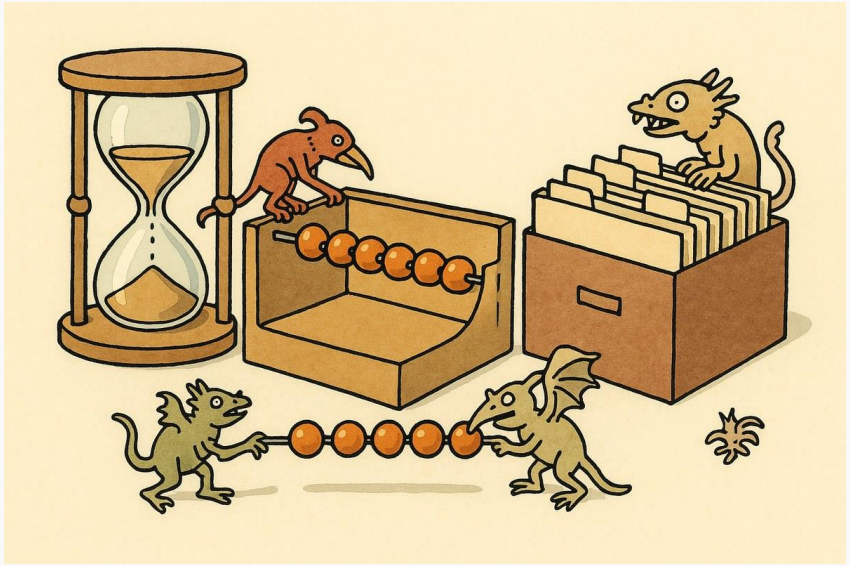
What kind of values: atomic types

- **Atomic**: we treat each as a single value in a cell
- numeric — 21, 1.75
- boolean — true or false
- string — "Cambridge"



What kind of values: compound types

- **Compound**: one value built from several
- **list** — several values in order
- **dict** — named values: a student record
- **text** — when the text is not just a label but has ‘components’ (semantic, syntactic, phonological, etc.)



Quantitative variables: discrete and continuous

- **discrete** — the possible values have '**gaps**': 0, 1, 2 siblings, never 1.5
- **continuous** — **in principle** any value in a range is possible: 1.7 m, 1.73 m, 1.732 m, etc.



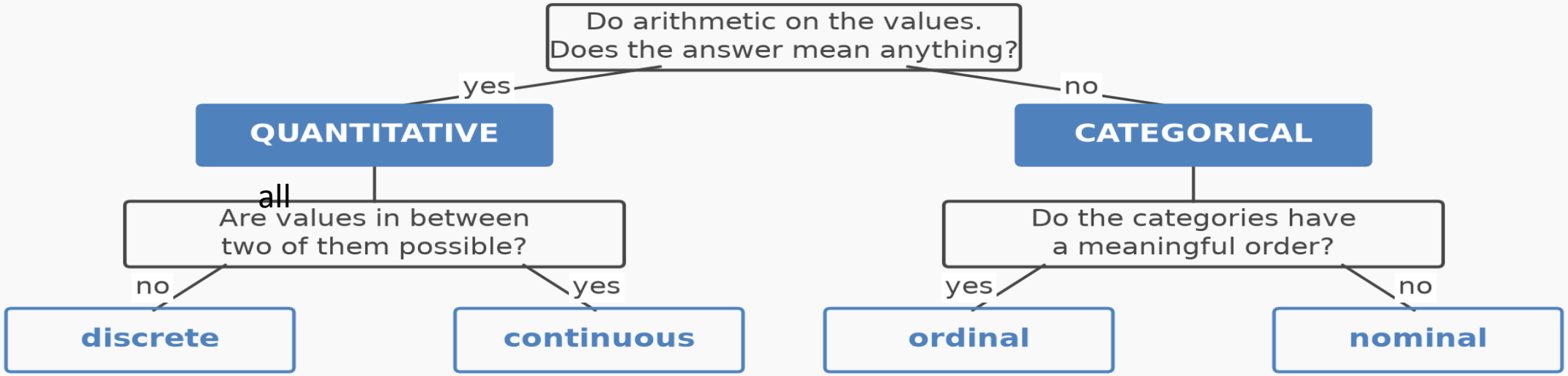
Categorical variables: nominal and ordinal

- **Categorical** — records which group something is in
- nominal — no order: **blood type, eye color, how you travel to campus**
- ordinal — a meaningful order: **shirt size S/M/L/XL**
- The test: **can you define an ordering that means something?**
- Ordered does *not* mean evenly spaced — L to XL need not be the same as S to M



Asking "what kind of variable is this?" first

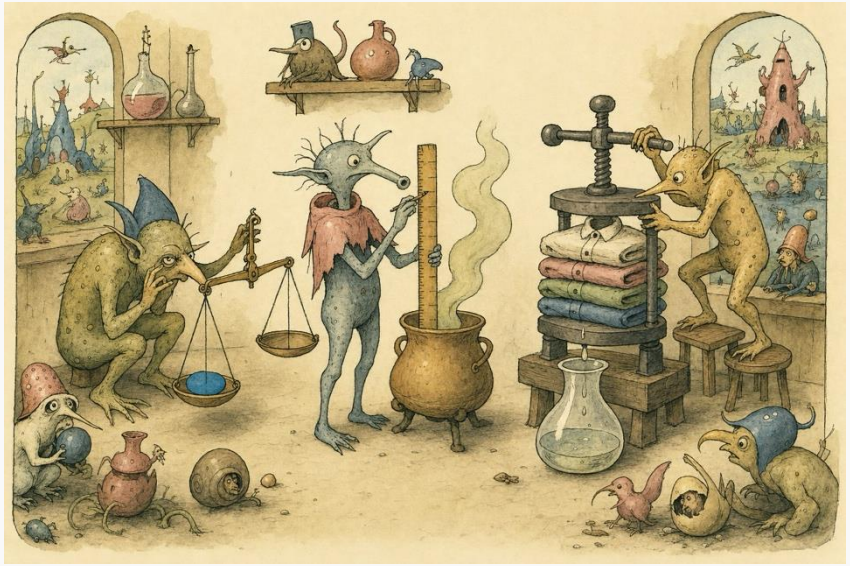
- First ask: **quantitative, ordinal, or nominal?**
- The type says **which summaries have a meaning**
- Ask ”what kind is this variable” before you try to compute anything!



A postcode is a number and is nominal — the arithmetic runs, and the answer means nothing.

Which summaries a variable allows

- **quantitative** — mean, median, standard deviation
- **ordinal** — median, mode, counts.
  - **A mean assumes the steps are equal…** are they?
- **nominal** — mode, counts, and proportions



The mean and the median

- **mean** — **add the values, divide by how many there are**
- **median** — sort them, take the middle one; with an even count, average the middle two

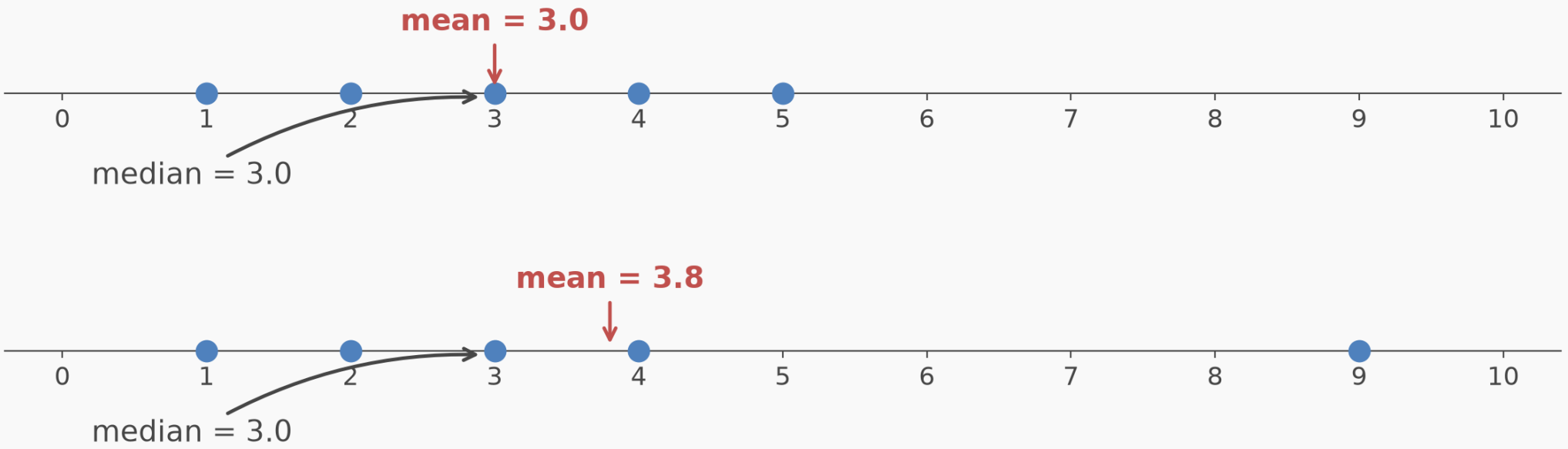
$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\bar{x} = \frac{3+5+5+8+14}{5} = 7.0$$

sorted: 3 5 **5** 8 14 median = 5.0

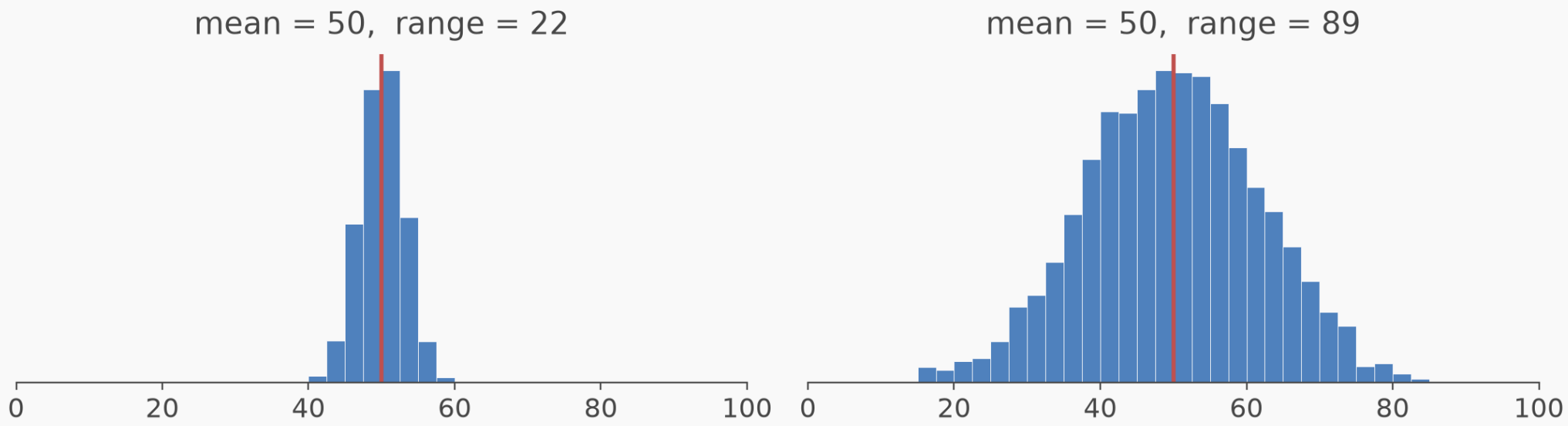
When the mean and the median disagree

- Same median. **One value moved.**
- The **tail** is the thin stretch of a few extreme values on one side
- **Skew is named for the side the tail is on**, *not* where most of the data sit



Spread: how far the values sit from the middle

- Two sets of numbers can have **the same mean and look nothing alike**
- **range — the largest value minus the smallest**



Variance and standard deviation

- **variance** (s^2) — the average squared distance from the mean
  - Why do we divided by **n – 1 ?**
- **standard deviation** (s) — **the square root of the variance**

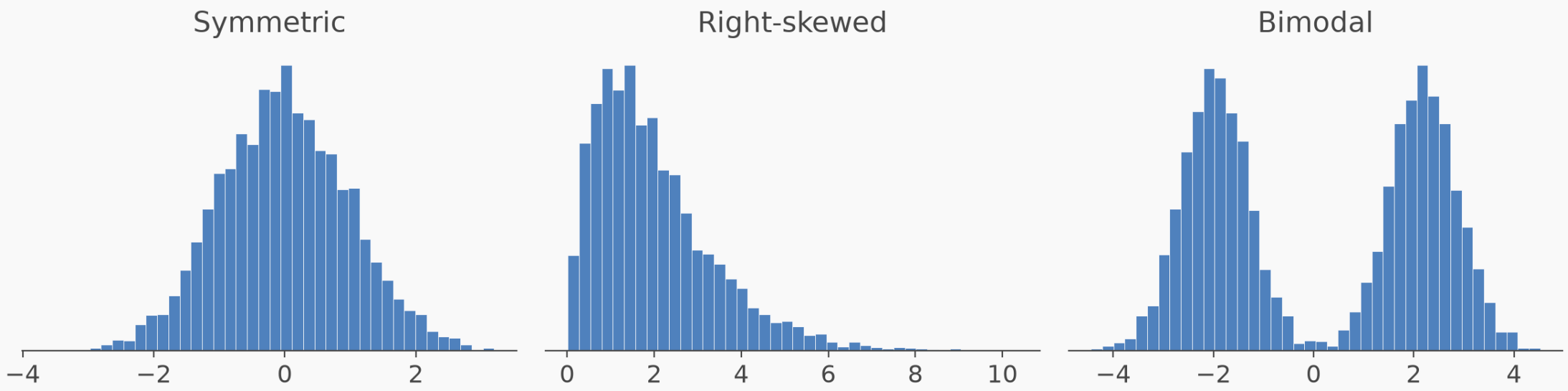
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

the variance is in squared units; the standard deviation is in the original units

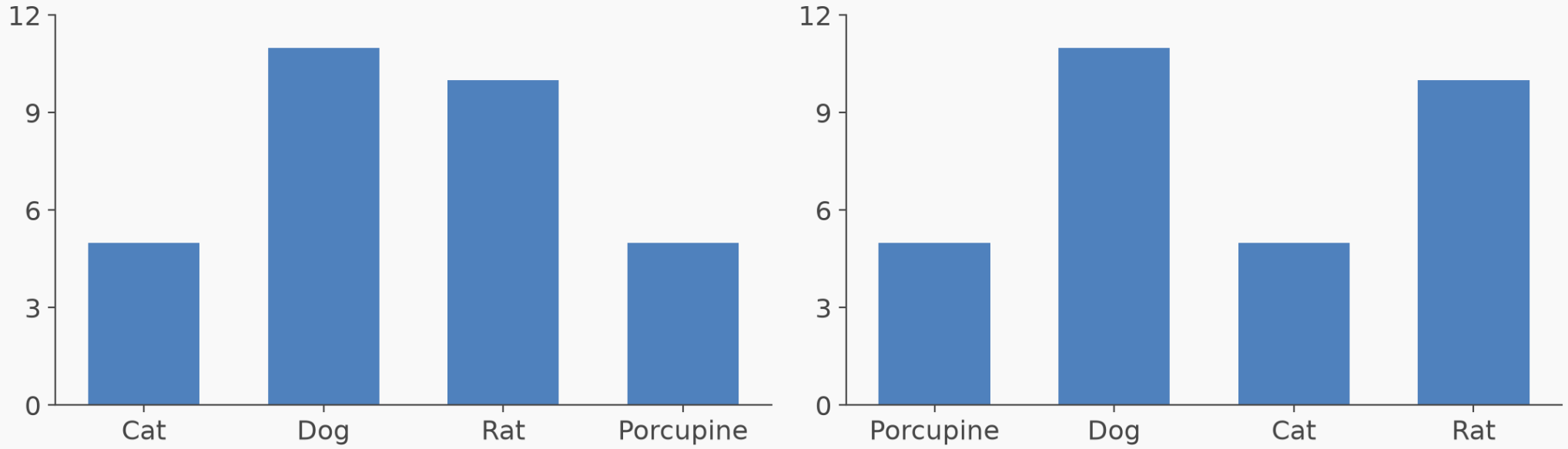
The distribution of a variable

- The values a variable takes, and **how often each occurs**



Describing a nominal variable

- Same data, two orders. **Neither is more correct.**
- **No skew, no symmetry** — these categories have no order so the placement is arbitrary!!
- mode — the most common category still makes sense



SECTION 4

Tables: messy and tidy

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Where data live: three kinds of storage

- **structured (tabular)** — fixed rows and columns: CSV, a database table, a Pandas DataFrame <3
- **semistructured** — keys and nesting, records may differ: JSON, XML, HTML
- **unstructured** — text, images, audio
- A web page returns HTML, and most APIs return JSON. **Turning either into rows** is often what we want.



Tabular data: records and variables

- Each **row** is one record — one thing observed
- Each **column** is one variable — the same fact about every row, blank where it was not recorded

In [4]:

Slide Type

```
imdb = pd.read_csv('imdb_top_1000.csv')
imdb.head()
```

Out[4]:

	Poster_Link	Series_Title	Released_Year	Certificate	Runtime	Genre	IMDB_Rating	Overview	Meta_score	Director	Star1
0	https://m.media-amazon.com/images/M/MV5BMDFkYT...	The Shawshank Redemption	1994	A	142 min	Drama	9.3	Two imprisoned men bond over a number of years...	80.0	Frank Darabont	Tim Robbins
1	https://m.media-amazon.com/images/M/MV5BM2MyNj...	The Godfather	1972	A	175 min	Crime, Drama	9.2	An organized crime dynasty's aging patriarch t...	100.0	Francis Ford Coppola	Marlon Brando
2	https://m.media-amazon.com/images/M/MV5BMTMxNT...	The Dark Knight	2008	UA	152 min	Action, Crime, Drama	9.0	When the menace known as the Joker wreaks havo...	84.0	Christopher Nolan	Christian Bale
3	https://m.media-amazon.com/images/M/MV5BMWMwMG...	The Godfather: Part II	1974	A	202 min	Crime, Drama	9.0	The early life and career of Vito Corleone	90.0	Francis Ford Coppola	Al Pacino

Words for the same idea

- row = observation · record · instance
- column = variable · attribute · feature · dimension
- sample — **the observations you actually collected**, out of all the ones you could have (or a subset of the data generally)

one column = one variable

title	year	minutes	colour
Arrival	2016	116	yes
Seven Samurai	1954	207	no
Spirited Away	2001	125	yes
Parasite	2019	132	yes

one row = one observation

Four problems to check a dataset for

- Values are **missing**
- Values are **wrong**
- The format is **messy**
- The data just **cannot answer your question** — and no amount of cleaning will fix it :'(



Three of those four, in one table

- The fourth one is not here — **you cannot see it in the data at all**

title	year	runtime
Arrival	2016	116
Jeanne Dielman	1975	way too long
(blank)	2001	about 2 hours
The Shawsh@nk Redemtpion	1994	142
Parasite	2019	2h

missing **wrong** **messy format**

Messy data: the produce-delivery table

- **What are the variables here?**

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

The same table in long form

- Nine rows, four columns
- **One delivery per row**
- **Every variable gets a column**
- **One kind of record per table**
- Next week's data?
  just append a new row!

ID	Time	Day	Number
1	Morning	Friday	15
2	Morning	Saturday	158
3	Morning	Sunday	10
4	Afternoon	Friday	2
5	Afternoon	Saturday	9
6	Afternoon	Sunday	20
7	Evening	Friday	55
8	Evening	Saturday	12
9	Evening	Sunday	45

Four things that make a table messy

- Headers are **values**, not variable names
- One variable in **both rows and columns**
- **Several variables** in one cell
- **Several kinds of record** in one table



Several kinds of record in one table

- Bigelow & Ducournau appear twice so their birth dates and film counts **repeat.**

one table

what does one row represent?

title	year	genre	mins	director	born	nationality	films
The Hitch-Hiker	1953	Film noir	71	Ida Lupino	1918-02-04	British-American	8
The Wandering Princess	1960	Drama	102	Kinuyo Tanaka	1909-11-29	Japanese	6
The Ascent	1977	Drama	111	Larisa Shepitko	1938-01-06	Soviet	5
Near Dark	1987	Horror	95	Kathryn Bigelow	1951-11-27	American	11
Point Break	1991	Action	122	Kathryn Bigelow	1951-11-27	American	11
Raw	2016	Body horror	99	Julia Ducournau	1983-11-18	French	3
Titane	2021	Body horror	108	Julia Ducournau	1983-11-18	French	3
The Substance	2024	Body horror	141	Coralie Fargeat	1976-11-24	French	2

one row per film

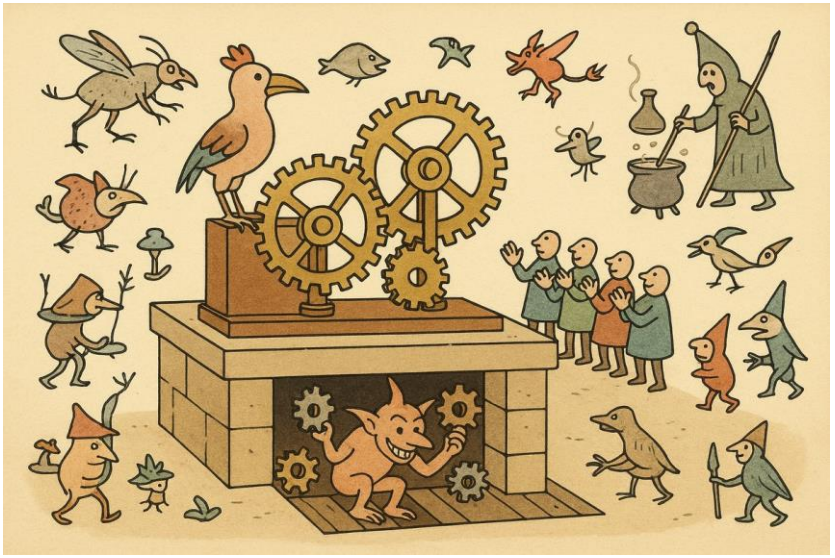
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one row per director

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SECTION 5

Live: student survey exploration (notebook)



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